⭐ LEVEL 2 — BASIC MCQs (FULL SET 1–120)

(We continue from **41 to 120**)

✔ MCQs 41–120

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41. Domain of f(x) = √(x+5) is:

a) $x > -5$
b) $x \geq -5$
c) $x \geq 5$
d) $x > 0$
Ans: b

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42. Domain of $f(x)=1/(x^2-9)$ is:

a) All x except ±3
b) All x
c) All $x > 3$
d) All $x < -3$
Ans: a

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43. Which function is one-one?

a) x^2
b) $|x|$
c) x^3
d) sinx
Ans: c

---

44. Which function is onto R?

a) x^2
b) e^x
c) x^3
d) $|x|$
Ans: c

---

45. A function is invertible if and only if it is:

a) One-one
b) Onto
c) Continuous
d) Bijective
Ans: d

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46. Range of $f(x)=|x|$ is:

a) R
b) $(-\infty,0)$
c) $[0,\infty)$
d) $(0,\infty)$
Ans: c

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47. Domain of $\log(x-2)$ is:

a) $x\geq 2$
b) $x>2$
c) $x<2$
d) All real
Ans: b

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48. Domain of $\sqrt{(4-x)}$ is:

a) $x\leq 4$
b) $x\geq 4$
c) $x<4$
d) $x>4$
Ans: a

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49. Range of $y = 1/(x^2+1)$:

a) $(0,1]$
b) $[0,1]$
c) $(0,\infty)$
d) $(1,\infty)$
Ans: a

---

50. $f(x)=x^2$ is invertible on:

a) R
b) $x\geq 0$
c) $x<0$

d) None
Ans: b

---

51. $f:R\to R$, $f(x)=x+5$ is:

a) One-one
b) Onto
c) Bijective
d) All
Ans: d

---

52. $f(x)=\sin x$ is invertible on:

a) R
b) $[0,2\pi]$
c) $[-\pi/2,\pi/2]$
d) $(0,\pi)$
Ans: c

---

53. $f(x)=\cos x$ is invertible on:

a) $[0,2\pi]$
b) $[0,\pi]$
c) $[-\pi/2,\pi/2]$
d) $(0,\pi/2)$
Ans: b

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54. Which is NOT a function?

a) Circle $x^2+y^2=4$
b) $y=x+1$
c) $y=x^3$
d) $y=1/x$
Ans: a

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55. Relation is a subset of:

a) A
b) B
c) $A\times B$
d) $A+B$
Ans: c

---

56. $f(x)=5$ is:

a) One-one
b) Constant
c) Onto
d) Invertible
Ans: b

---

57. Domain of tanx is:

a) All real
b) All real except odd multiples of $\pi/2$
c) $x>0$
d) $x<0$
Ans: b

---

58. Range of tanx is:

a) $[-1,1]$
b) All real
c) $(0,\infty)$
d) $[0,\infty)$
Ans: b

---

59. Range of cosx is:

a) R
b) $[-1,1]$
c) $(0,1)$
d) $[0,1]$
Ans: b

---

60. Range of sinx is:

a) $(-\infty,\infty)$
b) $(-1,1)$
c) $[-1,1]$
d) $[0,1]$
Ans: c

---

61. A function $f:A\rightarrow B$ must satisfy:

a) Every $a\in A$ has an image
b) Every $b\in B$ has a preimage

c) A must equal B
d) Domain = Codomain
Ans: a

---

62. Many-one function means:

a) Many outputs
b) Several different inputs give same output
c) One-to-one mapping
d) Bijective mapping
Ans: b

---

63. A function f:X→Y is onto if:

a) Range = Domain
b) Range = Codomain
c) Domain = Codomain
d) None
Ans: b

---

64. Number of functions from A (3 elements) to B (4 elements):

a) 64
b) 81
c) 12
d) 125
Ans: a ($4^3 = 64$)

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65. Cardinality of A×A if |A|=4:

a) 8
b) 12
c) 16
d) 32
Ans: c

---

66. If f is one-one and g is one-one, then g∘f is:

a) Not a function
b) One-one
c) Many-one
d) Onto
Ans: b

67. Inverse of f exists if f is:

a) Constant
b) Many-one
c) Bijective
d) Increasing
Ans: c

---

68. Range of secx is:

a) $[-1,1]$
b) $(-\infty,-1] \cup [1,\infty)$
c) $(0,\infty)$
d) None
Ans: b

---

69. $f(x)=x^3$ is:

a) Increasing
b) Decreasing
c) Constant
d) Periodic
Ans: a

---

70. $y=\sqrt{(x^2)}$ equals:

a) x
b) $-x$
c) $|x|$
d) $\sqrt{x}$
Ans: c

---

71. Range of e^x is:

a) $(0,\infty)$
b) $(-\infty,\infty)$
c) $[0,\infty)$
d) $[1,\infty)$
Ans: a

---

72. logx inverse is:

a) ex
b) x^2
c) 1/x
d) ln(x)
Ans: a

---

73. Relation "is father of" is:

a) Symmetric
b) Reflexive
c) Anti-symmetric
d) Equivalence
Ans: c

---

74. Relation "perpendicular to" is:

a) Not symmetric
b) Reflexive
c) Symmetric
d) None
Ans: c

---

75. All identity functions are:

a) Bijective
b) Injective
c) Surjective
d) All
Ans: d

---

76. $f(x)=|x-3| \geq$

a) 0
b) 1
c) 3
d) −1
Ans: a

---

77. Domain of $f(x)=1/|x|$ is:

a) $x \neq 0$
b) $x > 0$
c) $x < 0$

d) All real
Ans: a

---

78. Domain of $\sqrt{(x^2-4)}$ is:

a) $(-\infty,\infty)$
b) $x\leq-2$ or $x\geq2$
c) $x>2$
d) $x<-2$
Ans: b

---

79. $y = \sin^{-1}x$ is defined for:

a) All x
b) $x \geq 0$
c) $|x| \leq 1$
d) $x > 1$
Ans: c

---

80. Range of $\cos^{-1}x$ is:

a) $(-\pi/2,\pi/2)$
b) $[0,\pi]$
c) $[-\pi/2,\pi/2]$
d) $[0,2\pi]$
Ans: b

---

81. Range of $\tan^{-1}x$ is:

a) $(-\pi/2,\pi/2)$
b) $[0,\pi]$
c) $(0,\pi)$
d) $[0,2\pi]$
Ans: a

---

82. If $f(x)=x^3$, then $f(-x)=$

a) $f(x)$
b) $-f(x)$
c) $|x|$
d) None
Ans: b (odd)

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83. A reflexive relation must contain:

a) All (a,b)
b) Only (a,a)
c) (a,a) for all $a \in A$
d) None
Ans: c

---

84. x divides y is:

a) Reflexive
b) Transitive
c) Antisymmetric
d) All
Ans: d

---

85. Parity relation (“same parity”) is:

a) Not symmetric
b) Not reflexive
c) Equivalence
d) Partial order
Ans: c

---

86. $f:R \to R$, $f(x)=x^5$ is:

a) One-one
b) Onto
c) Bijective
d) All
Ans: d

---

87. $f(x)=x^2$ not invertible because:

a) Discontinuous
b) Many-one
c) Onto
d) Identity
Ans: b

---

88. $f(x)=\sin 2x$ is:

a) Periodic
b) Increasing

c) Decreasing
d) Constant
Ans: a

---

89. Composition is:

a) Always commutative
b) Never commutative
c) Sometimes
d) Undefined
Ans: c

---

90. Range of 1/x is:

a) All real
b) All real except 0
c) $(0,\infty)$
d) $(-\infty,0)$
Ans: b

---

91. If f(x)=0 for all x, it is:

a) One-one
b) Constant
c) Onto R
d) Bijective
Ans: b

---

92. tanx has period:

a) π
b) 2π
c) $\pi/2$
d) None
Ans: a

---

93. sinx has period:

a) π
b) 2π
c) $\pi/2$
d) None
Ans: b

94. If f is invertible, $f \circ f^{-1}$ is:

a) Zero function
b) Identity function
c) Constant
d) Undefined
Ans: b

95. Range of $|x-2|$:

a) R
b) $[0,\infty)$
c) $[-2,\infty)$
d) $(-\infty,0]$
Ans: b

96. Composite of constant and non-constant is:

a) Constant
b) Non-constant
c) Undefined
d) Zero
Ans: a

97. Domain of $f(x)=\sqrt{(2x-1)}$:

a) $x>1$
b) $x\geq1$
c) $x\geq1/2$
d) $x>1/2$
Ans: c

98. Range of $f(x)=x^2+4$:

a) $(4,\infty)$
b) $[4,\infty)$
c) $(-\infty,4)$
d) All real
Ans: b

99. Range of $|x|$ is:

a) R
b) $[0,\infty)$
c) $(-\infty,0]$
d) None
Ans: b

---

100. Domain of cotx:

a) $x \neq n\pi$
b) $x \neq (2n+1)\pi/2$
c) All real
d) $x>0$
Ans: a

---

101. f(x)=0/x is undefined at:

a) $x=0$
b) $x=1$
c) $x=-1$
d) No value
Ans: a

---

102. f(x)=secx undefined at:

a) $\pi/2$
b) π
c) $3\pi/4$
d) 0
Ans: a

---

103. f(x)=cscx undefined at:

a) $\pi/2$
b) π
c) 2π
d) $n\pi$
Ans: d

---

104. f(x)=|x| is:

a) Even
b) Odd
c) Neither

d) Bijective
Ans: a

---

105. $f(x)=x^3$ is:

a) Even
b) Odd
c) Constant
d) Periodic
Ans: b

---

106. $|x|$ has minimum at:

a) $x=0$
b) $x=1$
c) $x=-1$
d) None
Ans: a

---

107. Graph of odd function is symmetric about:

a) y-axis
b) x-axis
c) Origin
d) None
Ans: c

---

108. Graph of even function is symmetric about:

a) x-axis
b) y-axis
c) Origin
d) None
Ans: b

---

109. Range of $\tan^2 x$ is:

a) $[0,\infty)$
b) $(0,\infty)$
c) $(-1,1)$
d) R
Ans: a

---

110. Domain of f(x)=1/(x+3):

a) $x \neq 3$
b) $x \neq -3$
c) $x > -3$
d) All real
Ans: b

---

111. Number of one-one functions from A={1,2,3} to B={a,b,c}?

a) 6
b) 3
c) 9
d) 27
Ans: a (3!)

---

112. Number of relations on a set with 4 elements:

a) 16
b) 64
c) 256
d) 65536
Ans: d (2^{16})

---

113. Ceiling function gives:

a) Largest integer ≤ x
b) Smallest integer ≥ x
c) Fractional part
d) Integer part
Ans: b

---

114. $f(x)=e^x$ is:

a) One-one
b) Onto (0,∞)
c) Invertible
d) All
Ans: d

---

115. sinx is not invertible because it is:

a) Decreasing
b) Not one-one

c) Not continuous
d) Not defined
Ans: b

---

116. cosx is not invertible because:

a) Discontinuous
b) Not one-one
c) Not differentiable
d) None
Ans: b

---

117. tanx is invertible on:

a) $(-\pi/2, \pi/2)$
b) $(0, \pi)$
c) $[0, \pi]$
d) $(-\pi, \pi)$
Ans: a

---

118. 1/x is:

a) Even
b) Odd
c) Constant
d) None
Ans: b

---

119. $|x-3| \geq$:

a) 0
b) 3
c) -3
d) None
Ans: a

---

120. $f(x)=x^5$ is:

a) Bijective
b) Constant
c) Even
d) Periodic
Ans: a